Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	**A Real-time Detector of Driving Under the Influence (DUI) in Vehicle Environment using Multimodal Data**			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display (), Secondary Battery ( ), Advanced Mobility (√), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI (√), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1)]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Mechanical and Materials		Mechanical and Materials	
	Medium Category	Small Category	Medium Category	Small Category
	Automobile / Rail Vehicles	Safety Improvement Technologies	Automobile / Rail Vehicles	Vehicle Intelligence Technologies
Concept of Proposed Technology	**Problem & significance** Driving under the influence of alcohol and/or drugs remains a leading cause of traffic accidents, injuries and fatalities. In the U.S., as of 2023, over 30% of crash fatalities involve alcohol, translating to an estimated 34 deaths per day[1]. In the Netherlands, as of 2015, the number of road deaths			

[1] Drunk Driving | Statistics and Resources | NHTSA

due to alcohol use was estimated to be between 12-23%[2]. These are all preventable with effective detection and prevention measures. Conventional breathalyzers and ignition interlock devices (IID) can be helpful but are not suitable for continuous, non-intrusive prevention during normal driving (e.g., susceptible to spoofing, distracting, or socially intrusive), and they don't meet the evolving needs of automated vehicle take-over scenarios.

Regulatory & consumer testing context

Driver Monitoring Systems (DMS) that deliver Driver Drowsiness and Attention Warning (DDAW) are mandated in Europe (new types since 2024; all registrations since July 2026). Consumer testing further raises the bar: Euro NCAP 2026 introduces impairment (alcohol/drugs) detection into assessment protocols (interlock approaches are excluded), creating a clear market pull for non-contact DUI detection. This has not been actively developed yet, and requires further R&D effort for commercial application. This project aims to bridge that gap.

Proposed solution

There are three steps that need to be addressed:

1. Identify which biomarkers can be used to detect DUI, with high certainty and specificity.
2. Identify which non-contact sensors can be used to detect this.
3. Develop and validate an AI-based DMS prototype that has been trained based on the experimental data.

This project will conduct highly controlled, driving simulator experiments with people under the influence, and with ground truth measurements of their blood alcohol levels, as well as other key biomarkers to collect data. This will then be used to develop a DMS, in collaboration with the SME, that can detect DUI. This will be implemented on a vehicle based system, such that the latency is minimized.

Novel innovations

- Vision-based AI in Near Infrared (NIR) for robust 24/7 driver state: gaze (saccadic velocity, fixation, blink), facial/pose (area of interest) detection etc.
- Thermal (LWIR) + CAN analytics to profile lateral/longitudinal control, steering/judgment anomalies, vigilance & unusual maneuvers.
- Multimodal fusion that separates fatigue vs. intoxication (e.g., nystagmus patterns vs. slow eyelid closure) and distraction vs. intoxication (e.g., phone use vs. unusually slow gaze).
- Edge AI meeting GDPR via privacy-by-design and on-device processing.

[2] Alcohol - 4. How many casualties in the Netherlands are due to driving under the influence of alcohol?

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	Applications • Passenger cars seeking additional points in the Euro NCAP consumer testing for impairment detection. • Commercial vehicle fleets where there is an incentive for the fleet owner to ensure that DUI is not taking place.
Rationale for Joint R&D Project	Developing a reliable, non-contact system for detecting DUI requires expertise across physiology, human factors, driving behaviour, multimodal sensing, and AI. No single organisation can perform this work alone, as it involves the controlled induction of impairment, accurate physiological measurement, behavioural analysis, and real-time algorithm development. **Technical complexity** Detecting intoxication in naturalistic driving requires tightly controlled studies where impairment levels, physiological responses, and driving behaviour can be measured simultaneously. This demands an experimental environment that is safe, repeatable, and ethically governed, which cannot be achieved on public roads. **Complementarity of expertise** TNO contributes unique value through its independent research environment, ethics-compliant testing protocols, and advanced driving simulator facilities, enabling safe and scientifically valid data collection. TNO's long history in human factors, impaired-driving research, and physiological measurement ensures high-quality ground truth. The Korean SME complements this with sensor hardware, AI model development, and DMS integration capabilities, translating scientific insights into a deployable real-time system optimized for automotive ECUs. **Need for international collaboration** DUI detection is gaining regulatory relevance, especially with Euro NCAP 2026 impairment detection requirements, and Korean SMEs can benefit from access to European methodologies and datasets. Joint development enables demographically diverse data, broader applicability, and global market readiness.
Key Words	1. **Driver monitoring technology** to detect the alcohol/drug-induced abnormal state distinct from long/short distraction, phone use, microsleep, sleep, and unresponsive described in Euro NCAP 2026 2. **Multimodal Sensor Fusion** integrating Near-Infrared (NIR), thermal (LWIR) data streams, electrocardiogram (ECG) data, and Controller Area Network (CAN), which allows the system to cross-reference visible driver's states with invisible data

	3. **Near-infrared (NIR) vision-based AI** that keeps track of driver's physiological features including heart rate, facial expression, gaze (saccadic velocity, fixation duration, and blink dynamics), head pose, and behavior (action) 4. **Multi-modal AI** that keeps track of driver's driving patterns: lateral control, longitudinal control, steering & judgment, and vigilance & unusual maneuvers 5. **On-device AI** to process data locally on the vehicle's ECU (Neural Processing Unit), ensuring low latency and GDPR privacy compliance (no video leaves the car).

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	**Start of the Project**	Level (2)	**End of the Project**	Level (5)
R&D Objectives	The primary goal is to develop and validate a Driver Monitoring System (DMS) capable of detecting non-transient impairment (alcohol/drugs) to meet Euro NCAP 2026 standards. **Objective 1: Establish Ground Truth** • (Biomarker dataset) Scientifically validate the correlation between non-intrusive optical signals (e.g., pixel-based heart rate, gaze, head pose) and ground truth data (e.g., BAC levels or drug dosage, ECG, gaze, head pose) collected in a driving simulator. • (Driving dataset) In the driving simulator, collect data on the behaviour of the driver (in terms of acceleration, braking and steering inputs), with respect to lanes and other objects. • (Driving dataset): Scientifically validate the correlation between the ground truth data of driving behavior (e.g., in-lane behavior, car-following behavior etc.), with the level of intoxication (measured through the BAC levels or drug dosage). • Explore to what extent driver vehicle interaction changes by DUI. **Objective 2: Develop a Robust AI Pipeline** • Create a vision-based deep learning architecture to ensure 24/7 detection of driver impairment, fulfilling the DMS part of the Euro NCAP requirement for system availability in all lighting conditions. **Objective 3: Validate DUI detection** • Quantify sensitivity, and specificity for detection of DUI (e.g., distinction between intoxication and fatigue/distraction, where possible) by analysing patterns across multiple sensors and time. • Distinguish between the signatures of driving patterns with/without intoxication by analyzing multi-modal driving patterns.			

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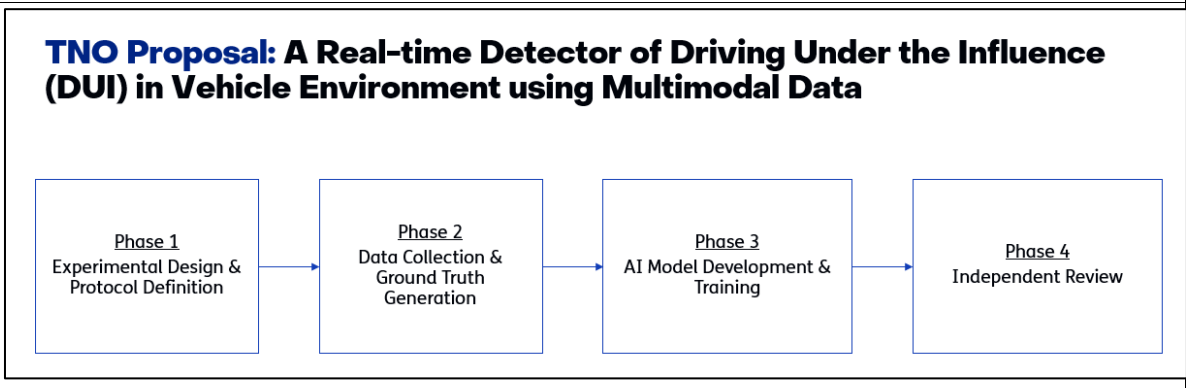
	Objective 4: Implement High-Performance Edge Computing • Optimize the complex multimodal AI models to run on a vehicle's embedded NPU (Neural Processing Unit) with low latency (<30ms) and ensuring that no biometric video data is stored or transmitted, complying with GDPR and privacy-by-design principles.
R&D Target Specifications	**Detection performance (alcohol/drugs):** ○ Target TPR (sensitivity) ≥ 90% in controlled simulator trials; ○ FPR ≤ 5–7%, ○ Tunable for market/brand policy. • **Differentiation performance:** ○ Impairment vs. fatigue/distraction separation accuracy ≥ 70% (balanced accuracy) in mixed scenarios. • **System performance:** ○ End-to-end latency ≤ 30 ms (edge NPU) ○ Availability over 85% ○ Privacy-by-design on-device processing (no cloud video).
R&D Plan	 **Phase 1: Experimental Design & Protocol Definition (Months 0–10, Lead: TNO)** • Regulatory mapping to Euro NCAP Safe Driving protocol; define Non-Transient State scenarios. • Ethical approval: secure medical ethics for alcohol/drug administration; define dosage, BAC measurement intervals, and safety procedures. • Synchronize all data streams (NIR/LWIR/ECG/CAN). • Sensor specification: SME defines placement, FoV, frame rate • Define and implement scenarios in driving simulator. Milestone M1: Approval from Medical Research Ethical Committee; simulator instrumentation ready; recruitment process defined. **Phase 2: Data Collection & Ground Truth Generation (Months 10–14 Lead: TNO)**

- Recruit participants based on the desired demographic distribution.
- Operate simulator; administer alcohol/drugs to participant, manage participants.
- Medical ground truth: record BAC/drug dose, ECG, skin temperature (LWIR) etc.
- Raw data capture: SME captures uncompressed sensor data for training.
- Data management and quality assurance.

Milestone M2: Curated biomarker and driving datasets with synchronized streams.

Phase 3: AI Model Development & Training (Months 14–22; Lead: SME)

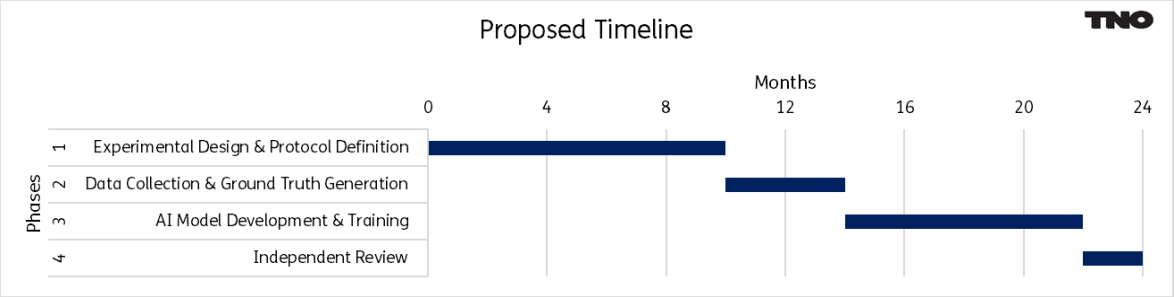
- Data transfer from TNO to SME (anonymized).
- Feature extraction: oculometrics (saccadic velocity, blink duration, pupil dynamics) and rPPG HR from video.
- Model training: correlate features with ground truth (e.g., frame ↔ BAC%).
- Multi-modal fusion model training: integrate optical + CAN for control anomalies; calibrate thresholds to minimize false positives.
- Model optimization for edge (ECU) deployment.

Milestone M3: Baseline models reach target validation metrics in simulator.

Phase 4: Independent Review (Months 22-24; Lead: TNO)

- TRL 5 demonstration and testing of the developed model in a small small-scale simulator study.
- The utilization of the training datasets, processes and validation conducted by the SME will be independently reviewed by TNO. A report summarizing the results of the review will be documented by TNO.

Milestone M4: TRL5 demonstration; documentation on independent review.



Evaluation & Measurement Plan

Primary metrics:

- Impairment detection TPR/FPR.
- Time-to-detection.

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	• Accuracy for impairment vs. fatigue/distraction. **Secondary metrics:** • Model latency • Availability (24/7, NIR/LWIR) • Privacy (no video offboard)

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	In this project, TNO is responsible for conducting the experiment, collecting the data and providing relevant human factors expertise. This includes programming and managing the driving simulator, recruitment and management of participants, and the collection of (ground truth as well as sensor) data. **Phase 1: Experimental Design & Protocol Definition** • **Regulatory Mapping:** Define specific test scenarios (e.g., "Non-Transient State" detection) based on the latest Euro NCAP Assessment Protocol. • **Ethical Approval:** Secure necessary medical and ethics committee approvals to administer alcohol and controlled substances to human subjects. • **Test protocol design:** Design the appropriate test protocol, and experimental standards. • **Program the simulator environment** with the appropriate scenarios and data synchronization etc. **Phase 2: Experiment Execution and Data Collection** • Facility Operation: Operate the simulator. • Subject Recruitment: Recruit a demographic mix of participants (age, gender, skin tone) to ensure AI fairness. • Collect medical ground truth information through dedicated sensors: • Record gaze behavior and other parameters, such as ECG etc. • Administer alcohol and measure BAC (Blood Alcohol Content) at fixed intervals. • Administer drug dose and measure the initial dose. **Phase 3: AI Model Development & Training** • Support interpretation of data. **Phase 4: Validation & Euro NCAP Pre-Compliance** • Documentation: Document a technical report documenting the outcomes of the validation study conducted.

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Expected R&R of Korean SME Partner	The SME is responsible for providing the DMS sensor hardware, developing the AI algorithms, and integrating as well as validating the solution.
	
	Phase 1: Experimental Design & Protocol Definition
	
	• Sensor Specification: Define the optimal placement, Field of View (FoV), resolution? and frame rates for the vision sensors within the simulator buck.
	• Data Requirement Spec: Provide Research Institute with a "Data Annotation Guide" specifying required variables (e.g., timestamp, measured heart rate, gaze direction and areas of interest etc.).
	
	Phase 2: Data Collection & Ground Truth Generation
	
	• Install the prototype camera modules into the simulator cabin.
	• Ensure the capture of raw, uncompressed sensor data for training.
	
	Phase 3: AI Model Development & Training
	
	• Feature Extraction: Develop computer vision pipelines to extract:
	  ○ Oculometrics: Saccadic velocity, blink duration, pupil dynamics.
	  ○ Vital Signs (rPPG): Remote heart rate estimation from video.
	• Model Training: Train Deep Neural Networks (DNN) to correlate the extracted features with the Ground Truth labels provided by research institute (e.g., "Video Frame A = 0.05% BAC").
	• Optimization: Compress the models (Quantization/Pruning) to run on low-power automotive NPUs.
	
	Phase 4: Validation & Euro NCAP Pre-Compliance
	
	• Fine-tune sensitivity thresholds based on research institute's validation feedback to minimize False Positives.

5. Expected Outcomes/Impact

Expected Outcomes and Impact	**Technical outcomes** • Validated DUI impairment detector (alcohol/drugs) with multimodal fusion and edge deployment. • Datasets & protocols for Euro NCAP 2026 pre-compliance. **Industrial & economic impact** • Differentiated safety feature for OEMs; supports regulatory trends and consumer testing priorities.

	• Potential reduction in DUI-related incidents and enhanced trust in automated driving handovers.

6. References

(1) Drunk Driving | Statistics and Resources | NHTSA

(2) Alcohol - 4. How many casualties in the Netherlands are due to driving under the influence of alcohol?